

## **Synopsis**

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### **Air Quality Prediction Using ML Techniques**

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**Problem Id: 15**

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Section : K

- **Problem Description:**

Air pollution is a major environmental and public health issue, especially in urban areas. Increasing levels of pollutants such as PM<sub>2.5</sub>, PM<sub>10</sub>, NO<sub>2</sub>, CO, and ozone can lead to severe health problems including respiratory diseases, cardiovascular issues, and reduced life expectancy. Monitoring and predicting air quality is therefore essential for taking preventive measures and informing the public.

The Air Quality Index (AQI) is a standardized measure used to indicate the level of air pollution and its potential impact on human health. However, AQI levels vary dynamically based on pollutant concentrations and meteorological conditions such as temperature, humidity, and wind speed.

This project aims to develop a machine learning-based system that can predict air quality categories (Good, Satisfactory, Moderate, Poor, Very Poor, Severe) using pollutant and weather data. By analyzing historical data and identifying patterns, the system can assist in early prediction of air quality levels. This can help government authorities, environmental agencies, and individuals take timely actions to reduce exposure and mitigate risks associated with air pollution.

- **Steps of Implementation:**

The implementation of the proposed system follows a complete machine learning pipeline. First, the dataset is collected from reliable sources containing pollutant concentrations and meteorological parameters such as temperature, humidity, and wind speed.

Next, data preprocessing is performed, which includes handling missing values, removing redundant features, and converting timestamp data into meaningful features such as month and hour. The Air Quality Index (AQI) categories are calculated based on Indian CPCB standards. Feature selection is then applied using correlation analysis to remove highly correlated features and SelectKBest to select the most relevant features with respect to the target variable.

After preprocessing, model selection is carried out by choosing suitable classification algorithms such as Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM).

The selected models are then trained and validated using the processed dataset. During this stage, the dataset is split into training and testing sets to evaluate model performance.

Finally, evaluation is performed using metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC curve. Based on the evaluation results, the best-performing model is selected as the final model for predicting air quality categories.

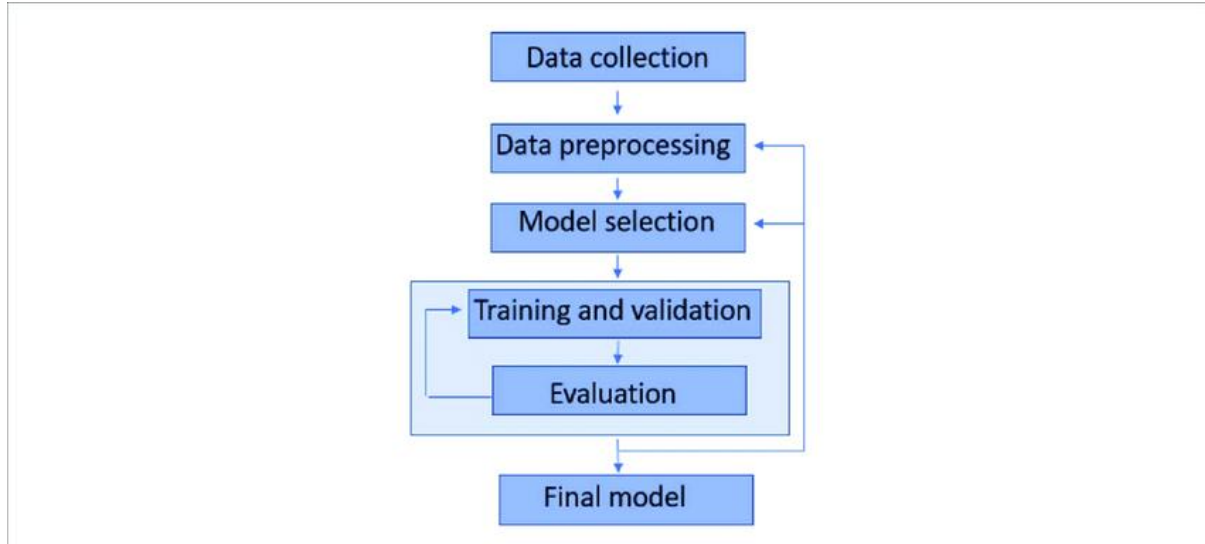


Fig. 1 Sample architecture

- **Expected Modules/ Libraries to be used :**

The following Python libraries will be used in the implementation of the project:

1. **pandas** – Used for data manipulation and analysis, including loading datasets, handling missing values, and performing data preprocessing operations.
2. **numpy** – Provides support for numerical computations and efficient handling of arrays and mathematical operations.
3. **matplotlib** – Used for basic data visualization such as plotting graphs and charts.
4. **seaborn** – Used for advanced statistical data visualization, including heatmaps and correlation plots.
5. **scikit-learn (sklearn)** – The primary machine learning library used for preprocessing, feature selection, model training, and evaluation. It includes:
  - Logistic Regression, KNN, and SVM classifiers
  - StandardScaler for feature scaling
  - LabelEncoder for encoding target labels
  - SelectKBest for feature selection
  - Train-test split and evaluation metrics

6. **datetime (inbuilt)** – Used for processing timestamp data and extracting features such as month, hour, and weekday.

- **Expected Evaluation metrics and Plots to be used:**

To evaluate the performance of the classification models, several standard evaluation metrics and visualization techniques will be used.

1. **Accuracy** – Measures the overall correctness of the model by calculating the proportion of correctly predicted instances out of the total. It provides a general idea of model performance.
2. **Precision** – Indicates how many of the predicted positive instances are actually correct. It is important when the cost of false positives is high.
3. **Recall** – Measures how many actual positive instances are correctly identified by the model. It is useful when missing important cases (false negatives) is critical.
4. **F1-Score** – The harmonic mean of precision and recall. It provides a balanced measure when there is class imbalance in the dataset.
5. **Confusion Matrix** – A tabular representation showing correct and incorrect predictions for each class. It helps in understanding which AQI categories are being misclassified.
6. **ROC Curve (Receiver Operating Characteristic Curve)** – Used to evaluate the model's ability to distinguish between classes by plotting true positive rate against false positive rate.
7. **Correlation Heatmap** – Used during preprocessing to visualize relationships between features and identify highly correlated (redundant) variables.

These metrics and plots together provide a comprehensive evaluation of the model's performance, ensuring both accuracy and reliability in predicting air quality categories.

- **Required topics from the subject:**

The following topics from the subject are used in this project:

- **Data Structures (Arrays/Lists)** – The dataset is stored and manipulated using arrays and lists (via pandas DataFrames). These structures are essential for handling large volumes of numerical data efficiently.
- **File Handling** – CSV files are used to store and load the dataset. File handling techniques are used to read, write, and manage data during preprocessing and model preparation.
- **Functions** – Functions are used to modularize the code, such as for AQI calculation, preprocessing steps, and feature transformation, making the code more organized and reusable.

- **Loops and Conditional Statements** – Used in data processing tasks such as applying AQI category rules, iterating over data, and implementing logical conditions for classification.
- **Basic Statistics** – Concepts such as mean, standard deviation, and correlation are used for data analysis, feature scaling, and identifying relationships between variables.
- **Object-Oriented Concepts (Basic)** – Libraries like scikit-learn use object-oriented programming where models are treated as objects with methods like fit() and predict().
- **Platform Required:**
- The project will be implemented using Python as the programming language. Jupyter Notebook will be used for data preprocessing, analysis, and model development due to its interactive and easy-to-use environment. GitHub will be used for version control and project management, allowing efficient tracking of changes and collaboration.
- **Books and Link Sources:** The following books and online resources will be referred to for this project:
- **Scikit-learn Documentation (Logistic Regression)**  
[https://scikit-learn.org/stable/modules/linear\\_model.html#logistic-regression](https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression)  
 This provides detailed information about Logistic Regression, its implementation, and usage in classification problems.
- **Scikit-learn Documentation (SelectKBest Feature Selection)**  
[https://scikit-learn.org/stable/modules/feature\\_selection.html#univariate-feature-selection](https://scikit-learn.org/stable/modules/feature_selection.html#univariate-feature-selection)  
 This source explains the SelectKBest method and ANOVA F-test used for selecting relevant features.
- **Central Pollution Control Board (CPCB) – National Air Quality Index (India)**  
<https://cpcb.nic.in/air-quality-index/>  
 This provides official guidelines for AQI calculation and classification according to Indian standards.
- **Kaggle – Air Quality Dataset**  
<https://www.kaggle.com/>  
 Used as a source for real-world air quality datasets for model development and testing.
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